
Experiment 9

Economic Dispatch in MATLAB

Lab Report

Student Name HAMZA AHMAD
Registration Number 2022-EE-165
Section B
Power System Operation and Control Laboratory
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MATLAB code for lossless and lossy ED:

```
clear; clc; close all;  
  
PLoad = [0; 0.217; 0.942; 0.478; 0.076; 0.112; 0; 0; 0.295; 0.09; 0.035;  
0.061; 0.135; 0.149];  
  
PD = sum(PLoad);  
  
fprintf('Total System Demand:\n');  
  
fprintf(' PD = %.4f pu (%.2f MW at 100 MVA base)\n\n', PD, PD*100);  
  
a_MW = [100; 120; 150; 180; 200];  
  
b_MW = [20; 25; 30; 35; 40];  
  
c_MW = [0.005; 0.010; 0.012; 0.015; 0.020];
```

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a = a_MW; b = b_MW * 100;      c = c_MW * 10000;

Pmin_MW = [50; 20; 15; 10; 10];
Pmax_MW = [200; 150; 180; 200; 200];

Pmin = Pmin_MW / 100; Pmax = Pmax_MW / 100;

n = length(a);
lambda = 2500;
max_iter = 100;
tol = 1e-6;
alpha_step = 50;

fprintf('Iter | Lambda ($/MWh) | ΣPG (pu) | ΣPG (MW) | Error (pu)\n');

for iter = 1:max_iter

    % C/P = lamda: b + 2cP = λ -> P =

    PG = (lambda - b) ./ (2*c);

    PG = max(Pmin, min(Pmax, PG));

    PG_total = sum(PG);

    error = PG_total - PD;

    if iter <= 5 || mod(iter, 10) == 0 || abs(error) < tol

        fprintf('%4d | %13.2f | %8.4f | %8.2f | %+9.6f\n', ...

            iter, lambda, PG_total, PG_total*100, error);

    end

    if abs(error) < tol

        fprintf('\n conv at itera %d\n', iter);

        break;

    end

    lambda = lambda - alpha_step * error;

end

C = zeros(n,1);
IC = zeros(n,1);

for i = 1:n

    C(i) = a(i) + b(i)*PG(i) + c(i)*PG(i)^2;

    IC(i) = b(i) + 2*c(i)*PG(i);

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end

C_total = sum(C);

fprintf('  E-D RESULTS\n');

fprintf('GENERATOR DISPATCH:\n');

fprintf('Gen | Output (pu) | Output (MW) | IC ($/MWh) | Cost ($/hr) \n');

for i = 1:n
    status = '';

    if abs(PG(i) - Pmin(i)) < 0.001
        status = '[at Pmin]';

    elseif abs(PG(i) - Pmax(i)) < 0.001
        status = '[at Pmax]';

    end

    fprintf(' G%d | %10.4f | %10.2f | %10.2f | %10.2f | %s\n', ...
        i, PG(i), PG(i)*100, IC(i), C(i), status);

end

fprintf('-----\n');

fprintf('TOTAL| %10.4f | %10.2f |           | %10.2f |\n', ...
    sum(PG), sum(PG)*100, C_total);

fprintf('-----\n\n');

fprintf('SUMMARY:\n');

fprintf('  System Lambda ( $\lambda$ ) = %.2f $/MWh\n', lambda);

fprintf('  Total Generation = %.4f pu (%.2f MW)\n', sum(PG), sum(PG)*100);

fprintf('  Total Load = %.4f pu (%.2f MW)\n', PD, PD*100);

fprintf('  Power Balance = %.6f pu (%.4f MW)\n', sum(PG)-PD,
    (sum(PG)-PD)*100);

fprintf('  Total Cost = $%.2f/hr\n\n', C_total);

```

RESULTS

Generator outputs and total cost (MATLAB). Cost curve plots with dispatch points.

Total System Demand:

PD = 2.5900 pu (259.00 MW at 100 MVA base)

Iter	Lambda (\$/MWh)	Σ PG (pu)	Σ PG (MW)	Error (pu)
1	2500.00	2.5500	255.00	-0.040000
2	2502.00	2.5500	255.00	-0.040000
3	2504.00	2.5500	255.00	-0.040000
4	2506.00	2.5500	255.00	-0.040000
5	2508.00	2.5500	255.00	-0.040000
10	2518.00	2.5500	255.00	-0.040000
20	2538.00	2.5500	255.00	-0.040000
30	2547.40	2.5870	258.70	-0.003003
40	2547.97	2.5898	258.98	-0.000169
50	2548.00	2.5900	259.00	-0.000010
58	2548.00	2.5900	259.00	-0.000001

conv at itera 58

Command Window

conv at itera 58

E-D RESULTS

GENERATOR DISPATCH:

Gen	Output (pu)	Output (MW)	IC (\$/MWh)	Cost (\$/hr)	
G1	2.0000	200.00	2200.00	4300.00	[at Pmax]
G2	0.2400	24.00	2548.00	725.76	
G3	0.1500	15.00	3036.00	602.70	[at Pmin]
G4	0.1000	10.00	3530.00	531.50	[at Pmin]
G5	0.1000	10.00	4040.00	602.00	[at Pmin]

TOTAL	2.5900	259.00		6761.96	
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SUMMARY:

System Lambda (λ) = 2548.00 \$/MWh
Total Generation = 2.5900 pu (259.00 MW)
Total Load = 2.5900 pu (259.00 MW)
Power Balance = -0.000001 pu (-0.0001 MW)
Total Cost = \$6761.96/hr